Final Project: Proposal

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Research Question

“Did worldwide Google searches for the phrase “end of the world” rise during or shortly after major news events?”

Newsrooms and media researchers might care about this because it could show how people respond to major events online. I expect the biggest increase to be around the start of COVID-19 in March 2020, with smaller increases around the October 2023 attack on Israel and the start of Russia’s full scale invasion of Ukraine. I would be surprised if the 2024 U.S. presidential election had a bigger increase than the pandemic.

Source and collection method

I will use the Google Trends Explore page, set to Worldwide and All Categories, and download the chart as a CSV file. Google says Trends data is sampled and scaled from 0 to 100. The scores show relative search interest, not the total number of searches (Google, n.d.-b). I will use methods from Chapters 5 (Protocols) and 6 (Static Pages) of the Web Data Science Textbook.

Scope and analysis

My downloaded file contains 70 monthly records, from March 2020 through December 2025. I will look at the months around four events: the World Health Organization’s announcement that COVID-19 could be characterized as a pandemic on March 11, 2020 (World Health Organization, 2020); Russia’s invasion of Ukraine on February 24, 2022 (United Nations, 2022); the October 7, 2023 attack on Israel (United Nations Human Rights Council, 2024); and the 2024 U.S. presidential election on November 5 (Federal Election Commission, 2024).

For each event, I will compare the event month and the following month with the three months before it. My file begins in March 2020, so it does not include the months before the pandemic. I can describe the March 2020 score, but I cannot compare it with a pre pandemic baseline using this file.

I will make a line chart of monthly search interest and a table comparing the event windows. The CSV records March 2020 as 100, the highest value in the file. I will look for increases around each event, while keeping in mind that the scores are relative and do not show the number of searches. I will use Python in a Jupyter Notebook to analyze and chart the CSV. With pandas, I will load the file, convert the Month column into dates, and calculate the average Google Trends score for the three months before each event and for the event month plus the following month. I will compare those averages to see whether search interest increased around the events.

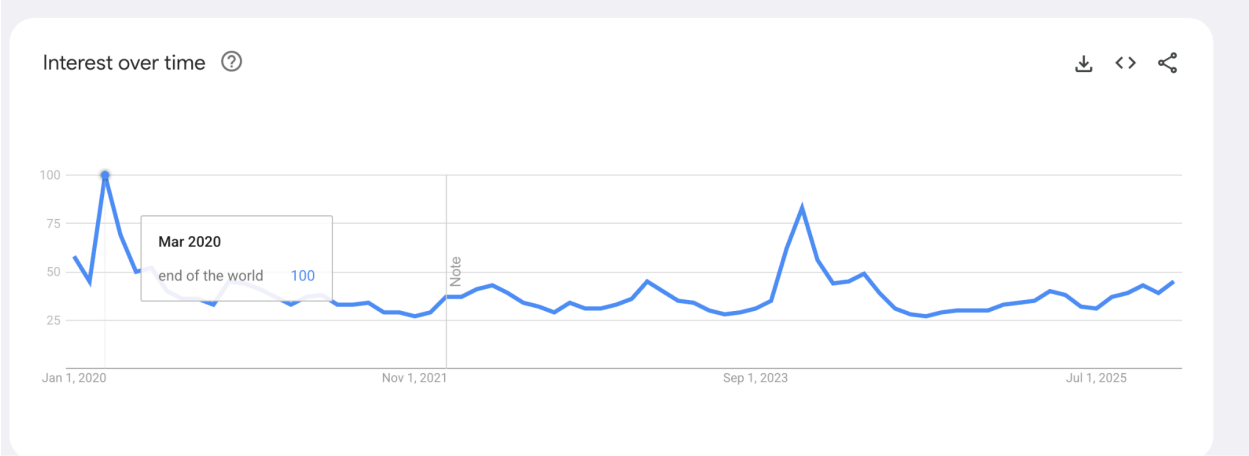
Ethics and Feasibility

Terms of service and robots.txt

I will collect the data using Google Trends' built-in CSV download, not by scraping pages or sending automated requests. Google provides an option to export chart data, and says that use of Trends information is subject to its Terms of Service (Google, n.d.-a, n.d.-c). I will credit Google Trends as the source. Since I am not crawling the site, I will not send repeated requests. If I later use automated collection, I will first check Google's terms and robots.txt instructions and follow applicable limits.

Personal information

I entered the search phrase “end of the world” on Google Trends and set the location to Worldwide. The Google Trends page at: https://trends.google.com/trends/explore?date=2020-01-01%202025-12-31&q=end%20of%20the%20world shows the query and settings. When I hovered over March 2020, the chart showed a score of 100. The matching row in my CSV was 2020-03,100. The fields I will use are Month and end of the world: (Worldwide). I used Google Trends to download the CSV, so I did not receive a separate HTTP status code.



The CSV does not include an HTTP status code, so I cannot report one from this download. The fields I will use are Month and end of the world: (Worldwide).

Biggest risk and fallback

The biggest risk is that Google Trends scores can change when the date range changes, and monthly data might miss a short spike depending on how I want to zoom into data. I will use one CSV export for all four event comparisons. I can still compare the Ukraine, Israel, and

election periods, and report March 2020 as the highest score in the file, but I won't claim searches rose around COVID compared with the months before it. I wanted to keep the search around post pandemic, however I might be open to changing the range to earlier if I find any events that might be prevalent.

AI disclosure

I used ChatGPT to discover sources that would be useful towards this project, reviewing terms of service, as well as revise my final proposal wording.

References

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United Nations. (2022, February 24). Noon briefing of 24 February 2022. https://www.un.org/sg/en/content/highlight/2022-02-24.html

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